

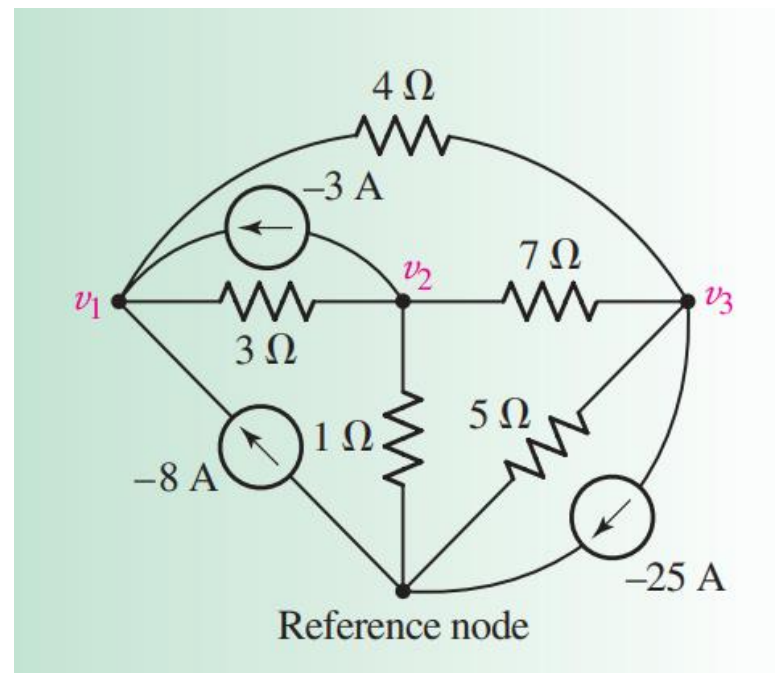
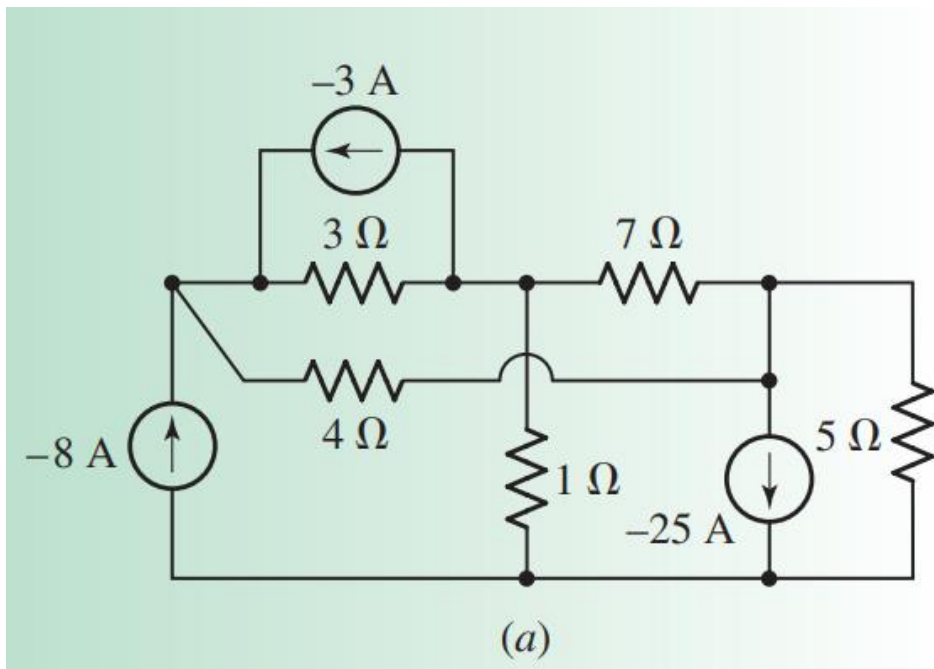
the number of nodes for which the voltage must be determined using nodal analysis is 1 less than the total number of nodes.

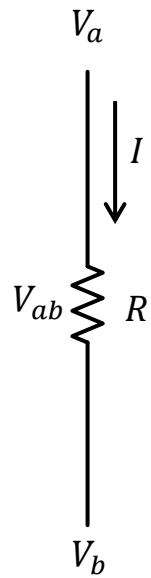
the number of equations required to solve for all the nodal voltages of a network is 1 less than the total number of independent nodes.

Nodal Analysis Procedure

- 1. Determine the number of nodes within the network.*
- 2. Pick a reference node, and label each remaining node with a subscripted value of voltage: V_1 , V_2 , and so on.*
- 3. Apply Kirchhoff's current law at each node except the reference. Assume that all unknown currents leave the node for each application of Kirchhoff's current law. In other words, for each node, don't be influenced by the direction that an unknown current for another node may have had. Each node is to be treated as a separate entity, independent of the application of Kirchhoff's current law to the other nodes.*
- 4. Solve the resulting equations for the nodal voltages.*

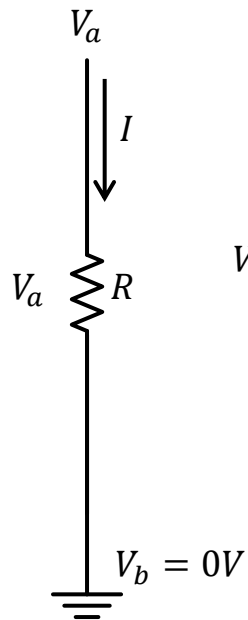
$$i = \frac{V_{\text{higher}} - V_{\text{lower}}}{R}$$



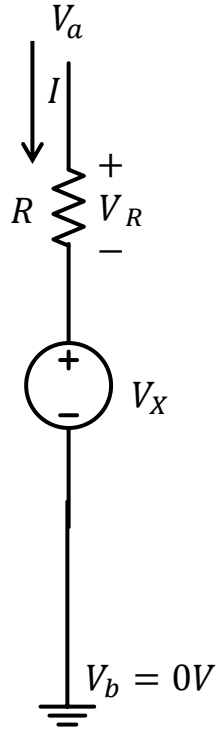


$$V_{ab} = IR$$

$$V_{ab} = V_a - V_b$$



$$V_{ab} = V_a - (V_b = 0) = V_a$$



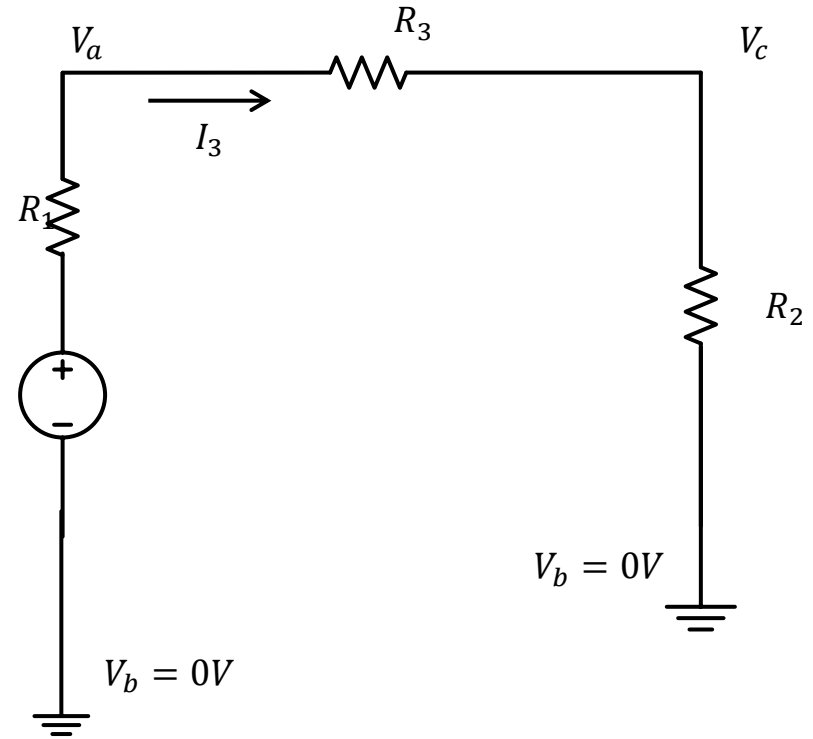
$$+V_X + V_R - V_a = 0$$

$$+V_X + IR - V_a = 0$$

$$I = \frac{V_a - V_X}{R}$$

$$+V_a - V_{R3} - V_c = 0$$

$$I_3 = \frac{V_a - V_c}{R_3}$$



Apply KVL from the node of your Interest (node a) to your neighbouring nodes(node c)

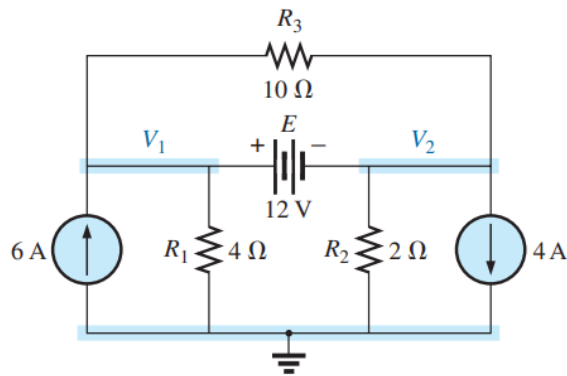


FIG. 8.62
Example 8.24.

A **supernode** is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.

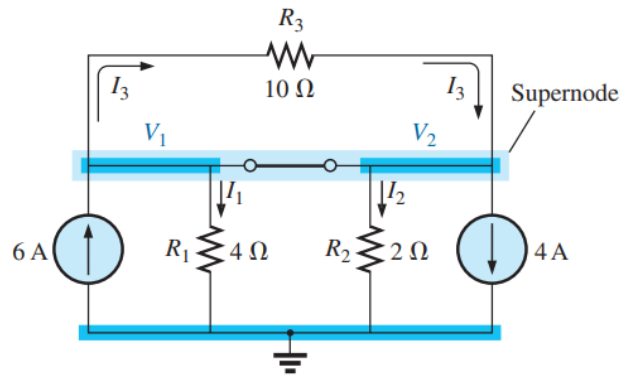


FIG. 8.63
Defining the supernode for the network in
Fig. 8.62.

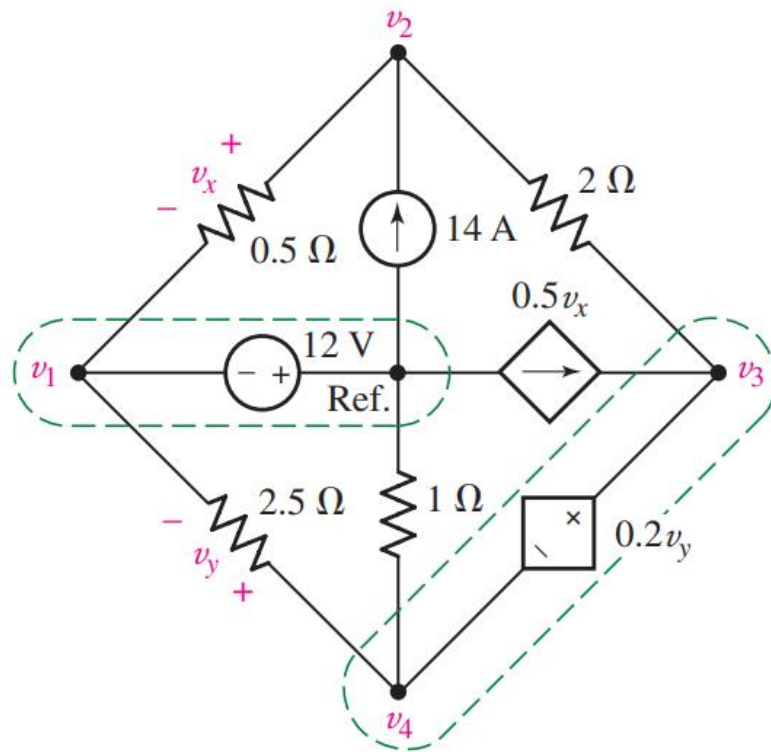


FIGURE 4.11 A five-node circuit with four different types of sources.

Determine the power supplied by the dependent source of Fig. 4.6a.

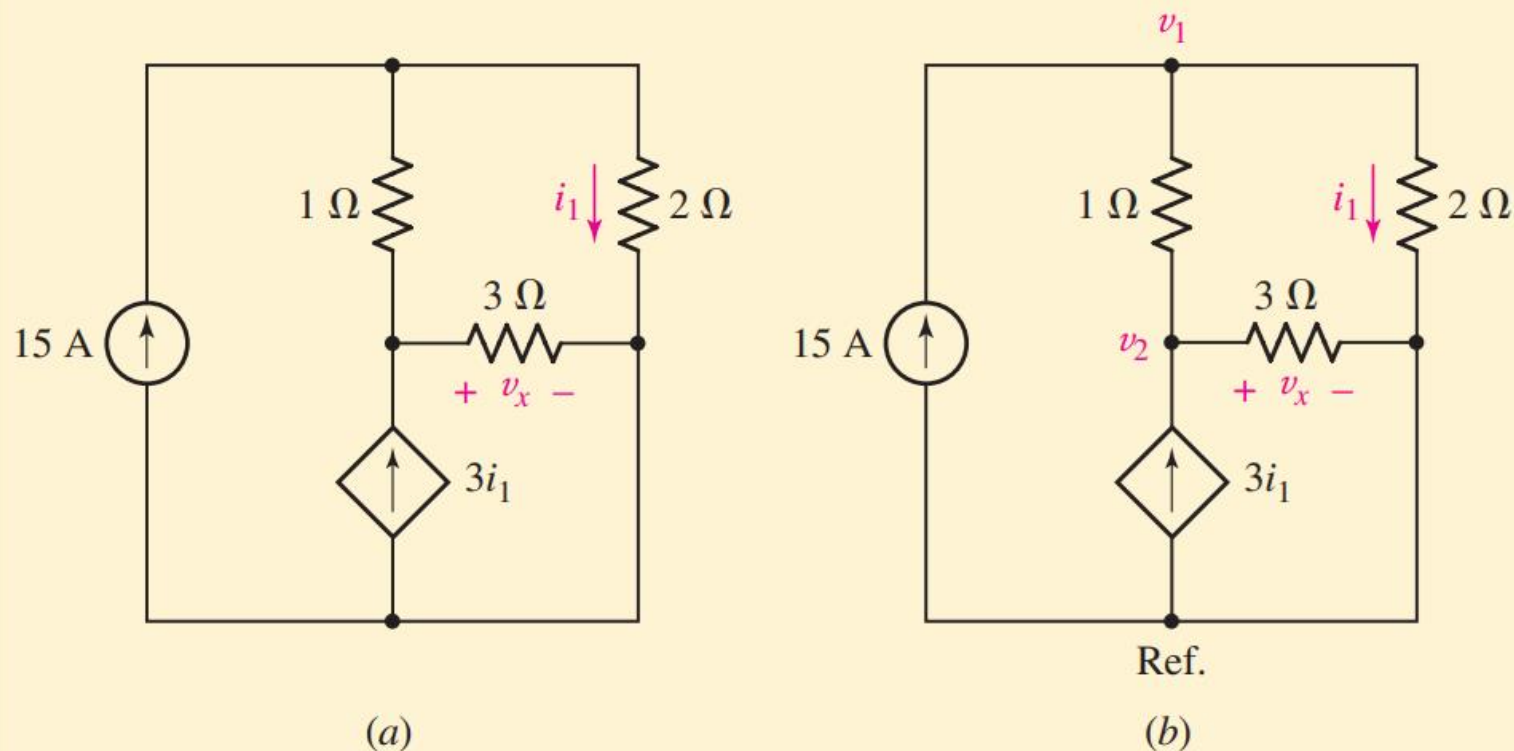


FIGURE 4.6 (a) A four-node circuit containing a dependent current source. (b) Circuit labeled for nodal analysis.